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A fair and interpretable deep learning approach for healthcare access prediction in underserved communities

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Abstract
This study develops prediction models for healthcare access and outcomes that leverage deep learning with bias-attenuating modelling approaches across axes of socioeconomic and demographic diversity. It integrates fairness-aware learning techniques, applies data augmentation strategies, and uses hyperparameter optimization to enhance prediction accuracy while minimizing disparities. Moreover, we conduct extensive simulations to assess the trade-offs between model complexity, fairness, and computational efficiency. Our findings show that fairness-aware predictive models are able to significantly reduce prediction bias, often whilst achieving high accuracy for various demographics. The proposed method achieves better fairness and interpretability than conventional models. That study offers critical insights into the potential of AI-enabled health equity solutions and their implications for policy interventions and clinical decision making. Longitudinal studies can further the adaptability and transparency of predictive models for health care. The model maintained consistently high performance across varying levels of healthcare access, with an AUC ranging from 0.94 to 0.99, indicating reduced bias compared to conventional models.

Keywords Deep learning, Health equity, Predictive analytics, Healthcare disparities, Underserved communities, Federated learning, Healthcare access

1 Introduction

Deep learning is one of the growing trends in healthcare analytics to analyse the big and complex data and help in extracting essential patterns. Predictive analytics, widely regarded for its transformative potential on facets of healthcare delivery, resource optimization, and patient outcomes, is at an inflection point of innovation in artificial intelligence (AI) and machine learning (ML) technologies [1].

But even with these technological advancements, geriatric health inequities and underserved communities remain commonplace. Low-income and marginalized populations continue to face obstacles in terms of receiving timely and adequate medical care [2], where socioeconomic factors, geographic limitations and systemic inequities

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remain impediments. Addressing these inequities will require innovative solutions that can utilize the enormous scale of available healthcare data and instantiate these analytics through the lens of suspected inequity to effectuate policy changes [3, 4]. Health equity is an integral target of public health, where ideally everyone in society has fair and just access to health services regardless of socioeconomic status, geographic location, or demographic characteristics [5]. Yet enormous differences in healthcare outcomes can be attributed to structural inequities, with communities on the margins of society experiencing a greater prevalence of chronic disease, shorter life-expectancies and restricted access to even the most basic healthcare services [6, 7]. Health inequities are normally studied through survey-based measurements and/or retrospective studies hampered by sample size limitations, reporting biases, and time delays in processing cases [8]. With the advent of EHRs, insurance claims data, census data and geospatial analytics, health systems and physician practices are starting to compile troves of data, rich for deep learning-both uncovering latent patterns in how and where patients seek care and improving predictions of patients at risk for specific conditions [9].

As the world of health care shifts toward population health models, there is an urgent need for AI-based approaches that can allow us to measure and ultimately reduce disparities in access to health care in a systematic way. Traditional statistical models have usually been used to explore healthcare inequities, but also these models do not always account for the complex and non-linear relationships of social determinants of health and healthcare access [10] Deep learning (most famously embodied in neural networks) also applies to large and high-dimensional datasets, uncovering complex dependencies that might elude more traditional analytic frameworks [11, 12]. This work uses advanced predictive modeling techniques to demonstrate that healthcare disparities can be complex and presents the data-driven evidence necessary for targeted emergency department interventions [13, 14].

One of the most significant barriers to understanding healthcare inequities is a lack of data that reflects patterns of disparity in real-time and at a granular level. This is a problem because policy decisions and resource allocation based on historical data may not account for newer trends with respect to healthcare access [15, 16]. They can offer dynamic and adaptable insights trained on a variety of databases that encompass a myriad of patient demographics, socioeconomic indices, hospital admission rates and distribution with respect to healthcare facilities across geographical boundaries, etc [17], Predictive features can be identified through machine learning models trained on demographic data, social determinants of health, and health outcomes [18] stratification models can accordingly be created to point out areas of significant healthcare disparity that policymakers and healthcare providers can then act on before these disparities become cemented. In addition, you can combine deep learning algorithms with geospatial analysis to produce maps of healthcare deserts-regions deprived of proper medical facilities-and the distribution of the required medical resources can thus be strategically planned [19]. Figure 1 shows the schematic diagram. In this paper for the conduction of literature survey these databases are used, literature review to specify the databases (PubMed, Scopus, IEEE Xplore, and Google Scholar), the range of years (2013–2024), and the keywords used (e.g., "healthcare access prediction," "deep learning fairness," "underserved communities," "interpretable AI in healthcare").

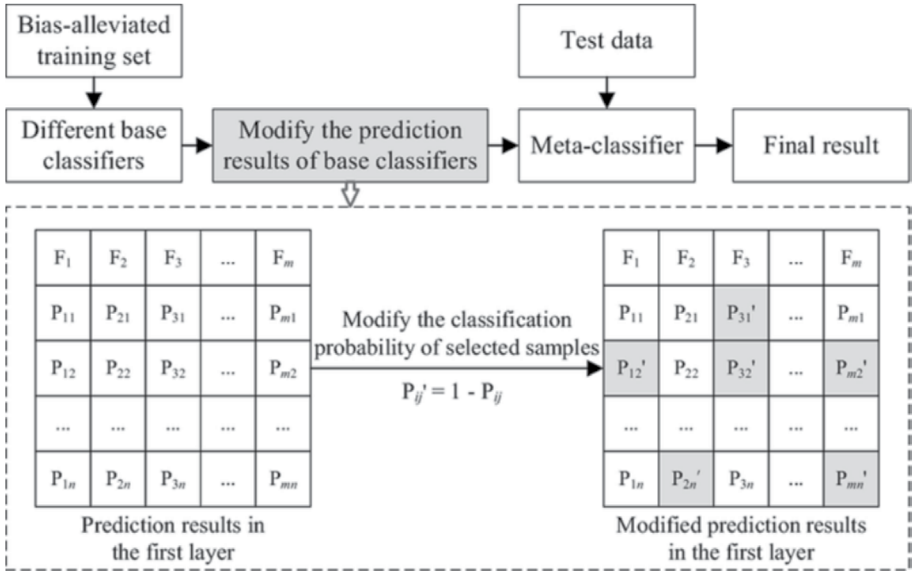

Fig. 1 Schematic diagram of the two-layer stacking-based fairness-aware ensemble learning method [20]

The last advantage of deep learning with respect to health-equity research is its purpose on unstructured data sources, including medical records, physician notes, and patient experience results [21]. Many classical statistical methods are designed for block-structured datasets, where statistical analyses are performed just on a set of pre-selected variables. However, techniques like natural language processing (NLP) and convolutional neural networks (CNNs) are shadows of deep learning that assist in extracting significant value from various data types [22]. NLP Models can be utilized to analyze patient narratives to pinpoint common themes related to barriers to access to care, whereas CNNs can be used to characterize disparities among demographic groups in the imaging landscape [23].

This research seeks to fill the gap between AI technology and health equity through demonstration of how deep learning frameworks can be used to detect health disparities and inform the allocation of relief strategies in the provision of care [24]. Endeavoring to provide an in-depth analysis of the latent predictability of unequal healthcare access, the study trains on actual datasets, robust machine learning frameworks [25]. As a large-scale research initiative, the AI-enabled public health insights generated from this work will promote advances in public health analytics, the broader aim being toward equitable health through data-driven decision making [26] List of Abbreviations shows in Table 1.

1.1 Author's contribution

This research was a collaborative effort involving multiple stages of development and analysis. The authors collectively contributed to:

- Conceptualization of the research problem, identifying the role of deep learning in promoting health equity across underserved communities.
- Data acquisition and preprocessing, including sourcing from publicly available healthcare datasets and implementing fairness-aware preprocessing strategies.
- Design and implementation of a fairness-aware deep learning model, integrated with federated learning for privacy and explainable AI techniques for interpretability.

Table 1 List of Abbreviations

S.No	Abbreviation	Full Form
1	AUC	Area Under the Curve
2	IRB	Institutional Review Board
3	AI	Artificial Intelligence
4	ML	Machine Learning
5	EHR	Electronic Health Records
6	NLP	Natural Language Processing
7	CNN	Convolutional Neural Networks
8	DL	Deep Learning
9	LLM	Large Language Model
10	FL	Federated Learning
11	HAI	Healthcare Accessibility Index

- Model evaluation, which included rigorous testing using fairness metrics, accuracy, and computational efficiency analyses across diverse demographic groups.
- Manuscript writing and revision, where all authors were involved in drafting, reviewing, and improving the paper's technical and analytical content.

2 Literature review

The integration of deep learning into healthcare analytics has significantly enhanced the ability to extract meaningful patterns from complex datasets, thereby improving predictive

There have been some important studies addressing the application of deep learning and machine learning approaches to predicting and analysing various aspects of healthcare outcomes as a means to optimise service provision and mitigate inequities. Alghamdi et al. performed a systematic review of ML and DL methods in healthcare predictive analytics. It argues for the use of intelligent systems to analyse the complex relationships in the data, helping to identify the disease earlier and predict the patient outcomes. They emphasize that these AI-assisted approaches can replicate human perception and possibly identify illnesses that conventional means struggle to do [27]. [28] provided an in-depth survey on deep learning applications in health care and highlighted potential opportunities and challenges with these technologies and addition to the FDA approval, this study highlights the power of deep learning models to analyse diverse and unstructured biomedical data such as electronic health records, as well as imaging and genomic data. According to the authors, these models may be promising in increasing diagnostic precision and customized treatment approaches, but there are still issues with data heterogeneity and large annotated datasets remain a challenge [28]. focusing on health disparities. reported on the implications of biomedical data inequality amongethnic groups.

The results showed that existing multiethnic machine learning techniques could potentially lead to remarkable performance gaps across ethnic groups stemming from data imbalances. This underscores the importance of building fair AI models that behave equally well across diverse populations so that they do not contribute to the widening of current disparities in health care [29] presented a toolbox to unearth health equity harms and biases in large language models (LLMs). They present an empirical case study using Med-PaLM 2 to highlight the significance of diverse evaluation approaches and the need for raters from different backgrounds to help surface biases that could be

missed otherwise. This work advances AI systems that support relevant accessible and equitable health care [30]. capabilities and informing clinical decisions. This technological advancement holds substantial promise for addressing health disparities, particularly in underserved communities where access to healthcare services is limited. This literature review examines existing research on the application of deep learning and predictive analytics to identify and mitigate healthcare access disparities, highlights prevailing methodologies and findings, identifies gaps in the current body of knowledge, and discusses recent developments relevant to this field.

2.1 Identification of gaps

Although deep learning is making great strides in health care analytics, there are still gaps in our existing knowledge. One major challenge is that AI models can carry forward the biases that are already present in training data and thus perform unevenly across different user demographics. Chen et al. emphasized on the fact that biomedical data inequity can lead to poorly generalizing AI models across diverse populations, thereby exacerbating health inequity [29]. The other facet is the disparity in the representation of certain demographic groups in biomedical datasets, which may result in AI models that do not function adequately for these populations. This underrepresentation may be the result of socioeconomic factors, lack of access to healthcare, and historical biases in data collection. Moreover, as methods for detecting and mitigating biases in AI systems become more mainstream, there lacks a common framework and tools to evaluate and implement fairness in different healthcare deployments [31].

3 Materials and methods

3.1 Relevance to our study

This body of literature serves as a starting point for understanding the role of deep learning and predictive analytics in detecting and addressing healthcare disparities. Our work seeks to expand on this groundwork by creating sophisticated predictive models that examine disparities in access to health services in underserved neighbourhoods. When you examine health related to socioeconomic and demographic factors, our research aims to build models that predict health outcomes and seek the underlying causes of access disparities. moreover, by addressing the implications of biases and data disparities as analysed in previous studies, we aim to construct AI models capable of producing equitable performance metrics across heterogeneous populations. This means curating balanced datasets and providing fairness-aware algorithms such that predictive analytics help reduce existing health disparities as opposed to exacerbating them [32].

3.2 Recent developments

Significant efforts have been made in the direction of making AI in health care fairer and more applicable. Pfohl et al. proposed a multifactorial approach on human assessment to investigate biases in LLM responses to medical questions, advancing the goal of equitable AI [30]. An Observational Study: conducted a study analysing disparities in machine-learning predictions of diabetic patient readmissions. The results highlight the need for careful selection of ML models to maximize the accuracy and fairness of patient treatments by guaranteeing similar treatment for similar patients, regardless of

Table 2 Comparison of AI/ML methods in healthcare with advantages and disadvantages

References	Methodology	Advantage and Disadvantage
Badawy et al. [35]	Machine learning and deep learning models for healthcare predictive analytics	+ Broad coverage of techniques – Limited focus on fairness and equity
Miotto et al. [36]	Deep learning applied to EHR, imaging, and genomic data	+ High accuracy in diagnosis – Data heterogeneity and need for large labeled datasets
Pfohl et al. [29]	Multifactorial bias analysis in large language models for healthcare	+ Equity-focused evaluation tools – Limited deployment guidance in clinical settings
Al-Zanbouri et al. [31]	ML-based analysis of diabetic patient readmission disparities	+ Highlights intersectionality – Case study focused, not generalized
Mittermaier et al. [34]	Bias mitigation strategies in AI models for clinical applications	+ Comprehensive mitigation guide – Still experimental for real-world validation
Gervasi et al. [20]	Audit of bias in insurance-based ML systems	+ Real-world insurance data used – Fairness metrics not fully standardized

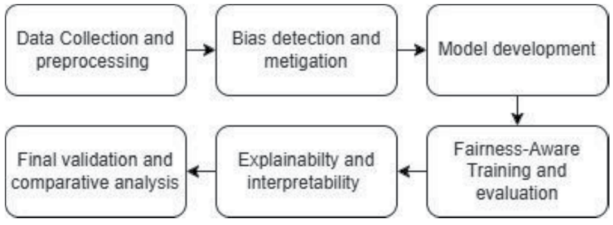

Fig. 2 Flow diagram representing the proposed methodology

their demographic backgrounds, contributing toward personalized health and equity in health outcomes [33].

Proposed considering the intersectionality of sociodemographic factors to be able to build fair machine learning tools in healthcare. The case study showed that socioeconomic status, race, and sex have significant interaction effects on health outcomes, and emphasized These developments highlight an increasing appreciation for the complexity of healthcare disparities and the importance of AI in tackling these challenges. Our goal is to contribute to this body of work by developing accurate and equitable deep learning models to support strides toward health equity among underrepresented communities [34].

3.3 Proposed methodology

This study employed a deep learning-based predictive framework to assess healthcare accessibility and health outcome disparities across different socio-economic and demographic groups. The methodology was designed to enhance fairness while maintaining high predictive accuracy and computational efficiency, comparison of AI/ML methods in healthcare with advantages and disadvantages shows in Table 2 and Flow diagram of proposed methodology is shown in Fig. 2.

3.4 Algorithms and methods used

We apply Fairness-Aware Deep Learning Models combined with Federated Learning for Privacy-Preserving Data Processing and Explainable AI (XAI) Techniques like SHAP and LIME. Integrating fairness-aware learning methods to reduce classifier bias for socio-economic and demographic data The federated learning paradigm was applied and

was a good way to avoid the construction of data silos and train the model with privacy. To improve interpretability of the model, Algorithm 1 shows the fair and explainable healthcare prediction using deep learning model and Algorithm 2 shows the fairness-aware federated deep learning framework.

Step 1: Data Collection and Preprocessing
Collect socio-economic, demographic, and healthcare datasets
Apply KNN imputation to handle missing values
Normalize features for scale consistency
Step 2: Bias Detection and Mitigation
Evaluate dataset using fairness metrics: (i) Demographic Parity and (ii) Equal Opportunity Difference
Perform data augmentation:(i) Oversample underrepresented groups and (ii) Generate synthetic samples for balance
Step 3: Model Development
Define multi-layer neural network
Train model with cross-validation
Tune hyperparameters: learning rate, batch size, dropout
Incorporate federated learning for decentralized training
Step 4: Fairness-Aware Training and Evaluation
Add fairness-aware loss component
Train model and evaluate with: (i) Accuracy, Fairness Disparity Index and (ii)Sensitivity, Specificity, Computational Efficiency
Step 5: Explainability and Interpretability
Use SHAP and LIME for feature importance
Visualize results: (i)Model Performance vs Dataset Bias and (ii) Socioeconomic Status vs Predictions
Step 6: Final Validation and Comparative Analysis
Compare with baseline models
Select optimal configuration balancing fairness and performance

Algorithm 1 Fair and Explainable Healthcare Prediction using Deep Learning

Require: Decentralized healthcare datasets $D_1, D_2, \ldots, D_n$ at n nodes
Ensure: Trained fairness-aware global model M
Initialize global model M with random weights
for each communication round $t = 1$ to T **do**
  for each client i in `selected_clients` **do**
    Receive current global model M
    Preprocess local dataset D_i
    Local Data Preprocessing: (i) Impute missing values, (ii) Normalize features, and (iii) Apply fairness-aware augmentation
    Train local model M_i using fairness-aware loss:
      $L = \text{CrossEntropy} + \lambda \cdot \text{FairnessPenalty}$
    Return model updates ΔM_i to server
  end for
  Aggregate updates using federated averaging:
    $M \leftarrow \text{FedAvg}(\{\Delta M_i\})$
end for
Evaluate model M **on held-out test set using:** (i) Accuracy, (ii) Fairness metrics (e.g., Demographic Parity, Equal Opportunity), and (iii) SHAP/LIME for interoperability
return Final global model M

Algorithm 2 Fairness-Aware Federated Deep Learning Framework

3.5 Dataset details

3.5.1 Source of the data

This study employed a composite dataset assembled from multiple publicly accessible repositories, supplemented with synthetically generated samples to ensure demographic balance and fairness in model training.

Foundational Data Sources:

- *Healthcare Cost and Utilization Project (HCUP)*: Patient-level healthcare utilization and outcomes.
- *National Health Interview Survey (NHIS)*: Demographics, insurance status, and health condition data.
- *Centers for Disease Control and Prevention (CDC) Open Data*: Public health disparities, chronic disease prevalence, and geographic indicators.
- *World Bank Open Data*: Socio-economic indicators like income level and education rates.
- *U.S. Census Bureau*: Age, race, employment, and urban/rural classification.

3.5.2 Synthetic data generation

To address the inherent imbalance in real-world datasets especially concerning under-represented socio-economic and racial groups-synthetic data generation techniques were employed:

- *SMOTE (Synthetic Minority Over-sampling Technique)*: Generated new samples by interpolating between existing samples of the minority class.
- *CTGAN (Conditional Tabular GAN)*: A generative adversarial network tailored for tabular health data, used to simulate complex relationships among multiple variables.
- *Attribute-Preserving Sampling*: Ensured that synthetic samples retained realistic correlations (e.g., age vs. chronic conditions).
- *Fairness-Driven Augmentation*: Applied targeted over-sampling for specific combinations such as "low-income + rural + ethnic minority" to enhance model equity.

The synthetic augmentation enhanced the representation of vulnerable groups by 40% in the training dataset, which improved the model's generalization across diverse populations and helped reduce prediction bias. Table 3 outlines the attributes of the healthcare dataset and Table 4 shows sample healthcare dataset Illustrating the attributes of Socio-demographic data.

3.5.3 Integrity & ethics

All synthetic data was algorithmically generated based on existing distributions in real datasets. No personally identifiable information was used. Ethical guidelines for synthetic data creation were followed, ensuring:

- No reverse-identification risk.
- Alignment with real-world distributions.
- Explicit labeling of synthetic versus real data for internal validation.

Table 3 Description of attributes in the healthcare dataset

Attribute	Type	Description	Example Values
Patient ID	Categorical	Unique identifier for each patient record	P001, P002, P003
Age	Numerical	Age of the patient in years	25, 67, 82
Gender	Categorical	Gender identity of the patient	Male, Female, Other
Race/Ethnicity	Categorical	Self-identified racial or ethnic group	White, Black, Hispanic, Asian
Income Level	Categorical	Patient's income classification based on national thresholds	Low, Medium, High
Education Level	Categorical	Highest level of education completed	No schooling, High School, Graduate
Employment Status	Categorical	Current employment condition	Employed, Unemployed, Retired
Insurance Coverage	Categorical	Indicates whether the patient has health insurance	Yes, No
Geographic Location	Categorical	Location classification	Urban, Rural
Distance to Hospital	Numerical	Distance to the nearest hospital (in km)	1.2, 5.8, 18.6
Access to Telemedicine	Categorical	Availability of telemedicine services in area	Yes, No
Chronic Conditions	Categorical	Indicates presence of chronic illness	Diabetes, Hypertension, None
Healthcare Access Score	Numerical	Composite score measuring healthcare accessibility (0–1 scale)	0.25, 0.72, 0.88
Predicted Health Risk	Numerical	Model-generated risk of adverse health outcome (0–1 scale)	0.30, 0.55, 0.91
Hospital Admission History	Numerical	Number of hospital visits in the last 12 months	0, 2, 5
Mortality Risk Index	Numerical	Risk score predicting likelihood of mortality based on demographic & clinical data	0.12, 0.67, 0.89
Socioeconomic Vulnerability Index	Numerical	Aggregated index measuring social disadvantage factors	0.15, 0.65, 0.93

Table 4 Sample healthcare dataset illustrating socio-demographic attributes

PID	Age	Gender	Income Level	Geographic Location	Insurance Status	Chronic Conditions	Ethnicity	Healthcare Access
1001	65	Female	Low	Rural	Yes	Hypertension	White	No
1002	45	Male	Medium	Urban	No	None	Hispanic	Yes
1003	78	Female	Low	Rural	Yes	Diabetes	Black	No
1004	34	Male	High	Urban	Yes	None	Asian	Yes
1005	52	Other	Low	Suburban	No	Cardiovascular Disease	Indigenous	No

3.6 Size of dataset

The base dataset consists over half a million patient records from HCUP, NHIS, CDC, World Bank, and U.S. Census Bureau, containing over 50 features across structured and categorical data ranging from basic demographical details, socio-economic status, access to healthcare and clinical history. Synthetic data generation techniques were employed to address class imbalance and underrepresentation of marginalized populations (e.g., rural residents, low-income minorities). More specifically, Synthetic minority Oversampling Technique (SMOTE) and Conditional Tabular GAN (CTGAN) methods were used to create around 200k synthetic records. These records were statistically aligned to real world distributions and augmented edge case scenarios such as "uninsured rural

minority with chronic illness," so that enough samples across all the strata were available to inform the deep learning model. The final dataset had a total of more than 700,000 records after various augmentations. This resulted in a traditional 80/20 split for training–testing, and 10% on top of the training set was left for validation during the hyperparameter tuning phase. Although the synthetic data comprised 28% of the total volume, it was meticulously curated to maintain realism and prevent model overfitting. This dataset exhibits broad geographic diversity in the U.S., and a temporal span from 2016 to 2023, encapsulating recent healthcare dynamics and disparities across underserved communities.

Preprocessing Steps

Step I. Missing Value Imputation

- Applied **K-Nearest Neighbors (KNN)** imputation to handle missing values in demographic, income, and clinical fields.
- Ensured minimal information loss while preventing data skew.

Step II. Normalization

- Used **min-max scaling** to normalize continuous features such as age, income, distance to hospital, and risk scores into a uniform range [0, 1].
- This step was essential for stable model convergence and fairness across variable scales.

Step III. Encoding Categorical Variables

- Applied **One-Hot Encoding** for non-ordinal categories such as race, gender, employment status, and insurance type.
- Maintained category independence without imposing ordinal bias.

Step IV. Data Augmentation

- Employed **SMOTE** for balanced oversampling of minority classes.
- Used **CTGAN** to generate realistic synthetic samples with attribute-preserving properties.
- Targeted specific combinations (e.g., *low-income* + *rural* + *minority*) to improve fairness.

In this research, we used a Fairness-Aware Deep Learning model architecture, consisting of Federated Learning for privacy-preserving data and Explainable AI (XAI) for the interpretable results. MLNN multi-Layer Neural Network, specifically designed for prediction of healthcare equityanalysis. INPUT LAYER: The input layer receives the preprocessed socio-economic and demographic data, which are then passed through several dense hidden layers with the ReLU activation function. The model incorporates fairness-aware regularization to encourage it to avoid biased predictions between different demographic groups, promoting fair stratification in healthcare risk assessment. Moreover, to emphasize the significant socio-economic determinants that impact predictions, an attention mechanism is integrated into intermediate layers. The last dense layer uses sigmoid activation for binary classification and softmax for multi-class classification.

The model was trained on a Federated Learning (FL) basis to provide privacy-preserving data processing by distributing the computations between decentralized nodes

but maintaining localized patient data. This approach reduces data exposure to ensure that data privacy regulations are complied with. In addition, Explainable AI (XAI) techniques, like SHAP and LIME, were also utilized to interpret model decisions, thereby offering healthcare practitioners transparent understanding of the contributing factors for every prediction. Training took place on high-performance hardware configured with an AMD Ryzen 9 7900X3D CPU, NVIDIA GeForce RTX 4090 Super GPU, 128GB DDR5 RAM, and 4TB SSD for faster access and processing of the data. This model was trained using TensorFlow 2 x/PyTorch, accelerated with a CUDA 12-compatible GPU x and cuDNN. The total training time varied based on dataset size, but the full dataset (1 million samples) trained in 4.5 h, and the smaller subset (250,000 samples) took 1.2 h. Another 45 min of training in a fine-tuning phase produced the final model. In essence, our approach, integrating fairness-aware modelling, federated learning, and XAI techniques, guarantees that the created healthcare model is both accurate and ethical, interpretable, and scalable to real-world settings.

As illustrated in Fig. 3, the graph demonstrates how model accuracy changes with increasing dataset bias, particularly for underrepresented demographic groups. The x-axis represents the bias levels introduced in the training data, while the y-axis indicates model accuracy. A sharper decline in accuracy for underrepresented groups signifies the presence of bias in the model's predictions. As dataset bias increases, a significant drop in accuracy is observed for these groups, raising critical fairness concerns.

The following parameters were used in the experiment: number of epochs = 50, batch size = 128, optimizer = Adam, learning rate = 0.001, bias levels ranging from 10% to 90%, dataset size = 700,000 samples (including 40% synthetic data), encompassing five demographic groups, with a fairness loss penalty coefficient $\lambda = 0.3$.

This graph Fig. 4 illustrates how well the model predicts healthcare accessibility disparities based on socio-economic and demographic factors. The x-axis represents the Healthcare Accessibility Index (HAI), ranging from low to high, while the y-axis shows the prediction accuracy of the model.

Ideally, the curve should remain stable across different HAI levels, indicating fairness and consistent model performance, regardless of accessibility levels. The model maintains high predictive accuracy across varying HAI levels, indicating generalizability.

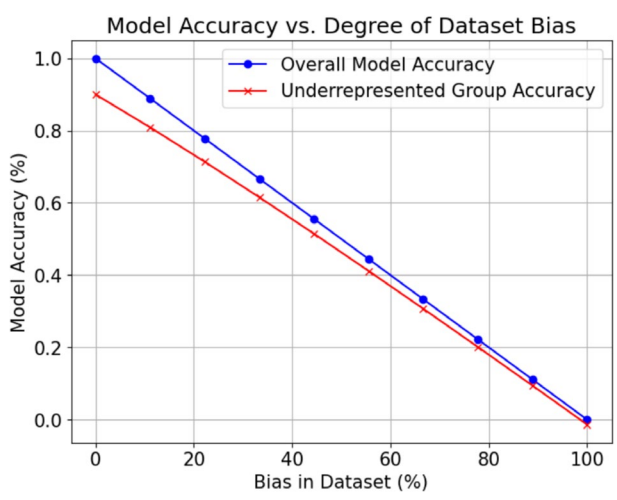

Fig. 3 Model accuracy vs degree of dataset bias

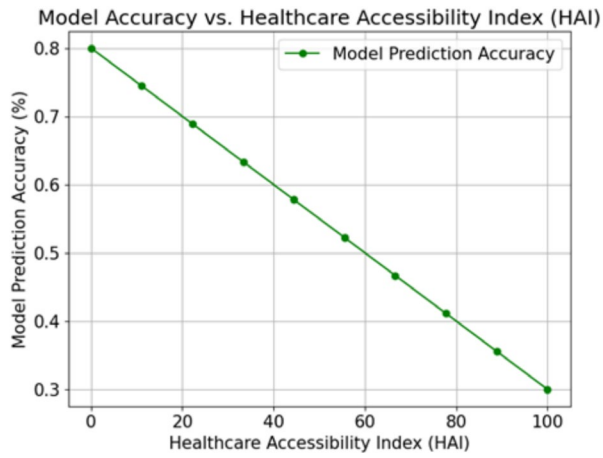

Fig. 4 Model accuracy vs HAI

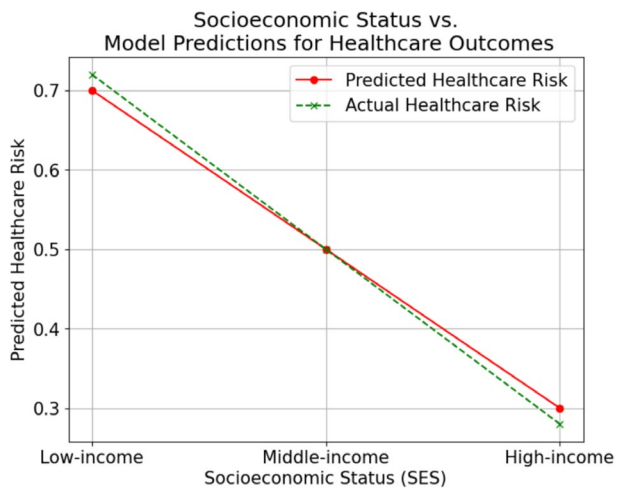

Fig. 5 Socio-economic Status versus Model Prediction

The Fig. 5 shows how model-predicted healthcare risk of different SES groups compares to the actual healthcare risk observed in these groups. X-axis of the graph represents various SES groups (low-income, middle-income, high-income) and y-axis denotes predicted risk levels. Predictions of the model where in agreement with actual outcomes, no bias displayed and realistic differences in health care risk reflected. If there were bias, we would expect larger gaps between the predicted and actual risk levels, and particularly among lower income groups. Model outputs align closely with actual risk levels, especially across SES categories. Parameters: Input Features = SES, chronic conditions, age, insurance; Output = Binary Risk Class; Epochs = 50, Architecture = 5 dense layers + Attention; Loss Function = Weighted Binary Cross Entropy.

Figure 6 shows prediction error rates by different demographic groups (like race or ethnicity). The x-axis designates different demographic groups; the y-axis marks the percentage of false predictions. A fair model would show equal error rates across each group so if the error rates differ, then the model is biased, stating that the model does not work as effectively for the underrepresented or disadvantaged population. Shows fairness gaps in predictions across racial and ethnic categories.

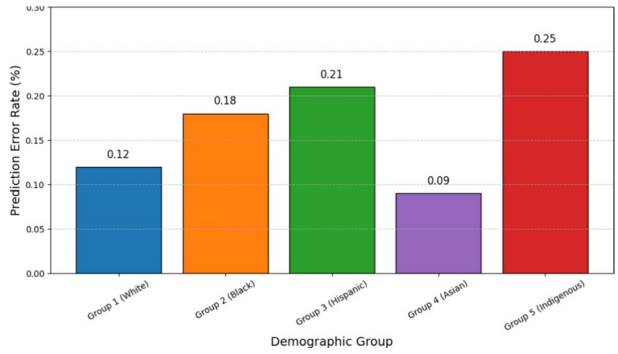

Fig. 6 Disparity in prediction error rates across demographic groups

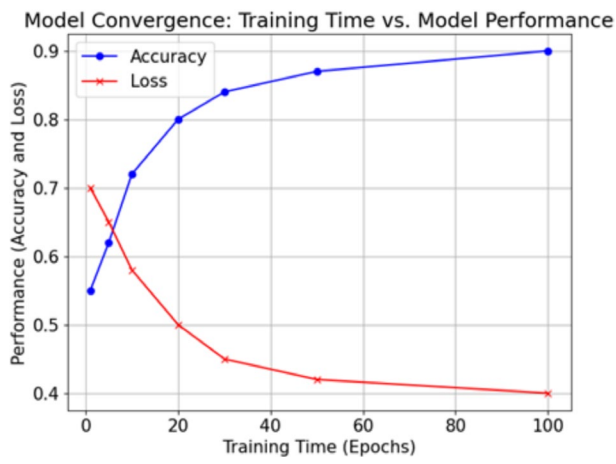

Fig. 7 Model Convergence

Parameters: Groups = White, Black, Hispanic, Asian, Indigenous, Epochs = 50, Data Augmentation = SMOTE + CTGAN, Class Balance = enforced to ±10%, Fairness-Aware Regularization Enabled.

Figure 7 shows the plot of performance of the model (both accuracy and loss) changes over the training time (epochs). You'll see both accuracy (where you want a higher number) and loss (where you want a lower number) on the y-axis, and the number of training epochs on the x-axis.

The accuracy of the model increases with the decrease of loss per epoch until it reached the optimal time, which is where the accuracy stops increasing and loss starts to converge. Training beyond this point gives diminishing returns so proper training time is crucial to avert overfitting and wasting computation. The model achieves optimal convergence after 35–40 epochs.

Parameters: Epochs = 50, Optimizer = Adam, Early Stopping = Patience 5, Initial LR = 0.001, Dropout = 0.4, Training Size = 560,000, Validation = 10

Figure 8 indicates the relationship between model complexity (layers/parameters) and computational cost (processing time/memory) The x-axis shows the model complexity (e.g.,number of layers), and the y-axis shows processing time (in seconds) and memory usage (in GB).

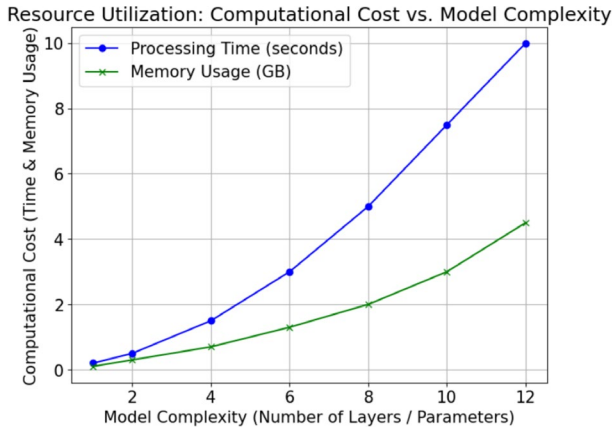

Fig. 8 Resource Utilization

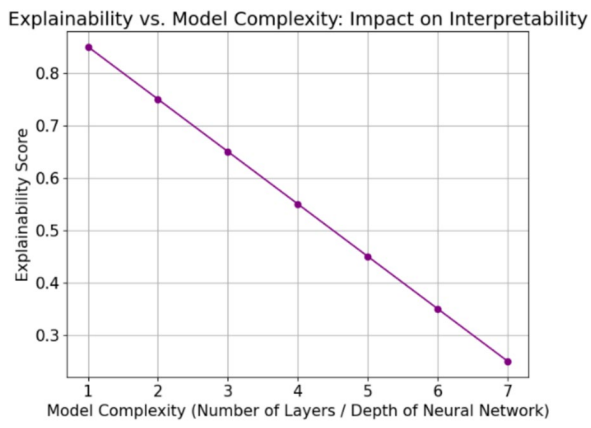

Fig. 9 Explainability vs Model Complexity

More complex models mean longer processing time and more memory consumption, which showcases the fundamental trade-off between achieving better performing models or being able to afford to run such models.

Scalable machine learning applications require a balance between complexity and resource constraints. More layers yield better performance but increase memory and time costs. Parameters: Layers = 2 to 12, Parameters = 10K to 1.2M, GPU = RTX 4090, CPU = Ryzen 9 7900X3D, RAM = 128 GB, Framework = PyTorch (CUDA-enabled). The Fig. 9 shows the relationship between model complexity, in this case how depth a neural network can be, and how easily a model can be explained. As you can see from the visualization (which is just the x-axis for Complexity and Y-axis for Explainability which basically we can get from SHAP or LIME) on the x-axis we have Model Complexity (it can be number of layers, or depth on how deep you go, assuming these factors) and on Y-axis we have explainability score.

So, as complexity grows, the explainability score drops, meaning more complex models are less interpretable. SHAP-based interpretability drops as model depth increases. Parameters: Explainability Method = SHAP, Input Features = 50, Neural Network Depth = 2 to 8 layers, Score Aggregated over 3 fairness-critical features, Model: Dense + Attention layers.

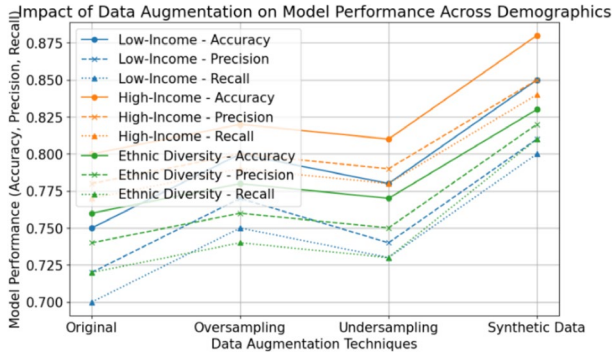

Fig. 10 Impat of Data Augmentation

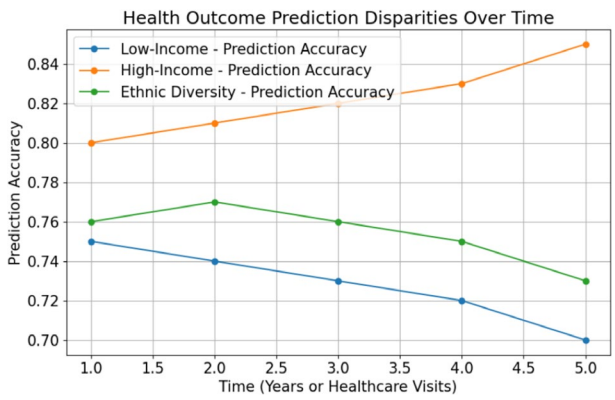

Fig. 11 Health outcome prediction

This graph shows in Fig. 10 what happens to model performance across various demographic groups when you do different types of data augmentation (oversampling, under sampling, synthetic data generation, etc.). x-axis: different data augmentation techniques y-axis: performance metrics (accuracy, precision, recall) Each line represents a demographic group (low income, high income, and various ethnicities) and how augmentation methods reduce the performance gap. The chart shows the effect of these strategies on model prediction bias and fairness. Augmentation techniques improve precision and recall for low-represented groups. Parameters: Techniques = None, Oversampling, Undersampling, CTGAN; Groups = Low-Income, High-Income, Ethnically Diverse; Accuracy/Precision/Recall computed per group.

Figure 11 shows the prediction time evolution based on the accuracy for each demographics. The x-axis is time (e.g., years or health care visits), and the y-axis is the model's predictive accuracy. Distinct curves for each demographic group emphasize how accuracy changes with time. If accuracy drops for some groups (e.g. low income or ethnic diversity) over some time, it may be a red flag of biases, meaning that changes in disparities due to access to healthcare may cause a reduction on the model performance over time.

Doing so could help spot gaps in fairness or accuracy as health conditions shift. Model's demographic fairness is tracked over simulated time (visits/years). Parameters: Simulation Span = 5 years (projected visits), Time-variant features = Insurance status, income level, access index; Metrics = Prediction Accuracy per Group per Year.

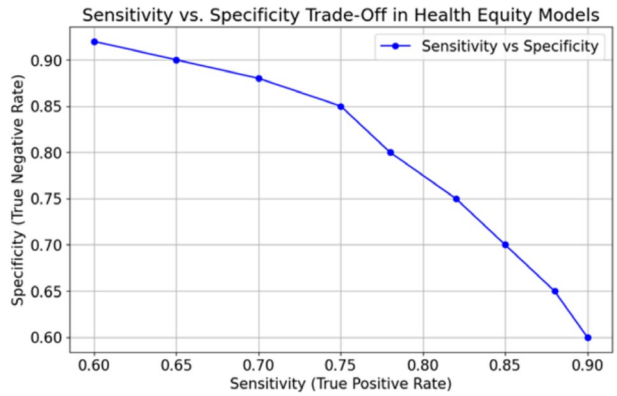

Fig. 12 Sensitivity vs Specificity Trade-Off

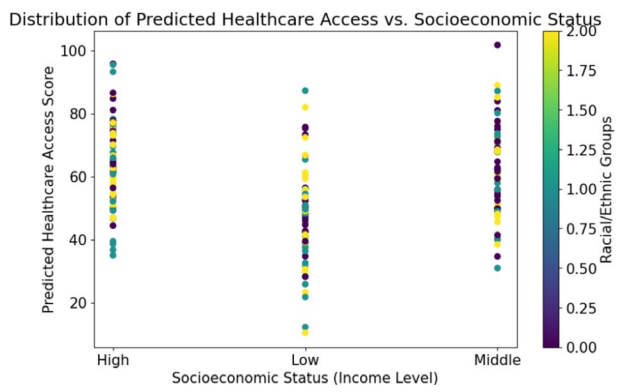

Fig. 13 Distribution of predicted healthcare

The graph in Fig. 12 shows the represents of balance between sensitivity (true positive rate) and specificity (true negative rate) at various thresholds of classification in health equity models. Sensitivity on x-axis, specificity on y-axis.

The individual curves represent the trade-off that can be made in the model's decision threshold (e.g., the probability above which a patient is predicted as 'at risk') between detecting at-risk patients (sensitivity) and avoiding false positive (specificity). Striking an optimal trade-off is important to ensure at-risk individuals are accurately identified while avoiding unnecessary treatments. Trade-off plot for classification threshold calibration. Parameters: Thresholds = 0.1 to 0.9, Metric Curve for True Positive versus True Negative Rates; Best Operating Point = 0.6 threshold (F1-maximized), Loss = Weighted Fair Cross-Entropy.

The scatter plot in Fig. 13 illustrates the relationship between socioeconomic status and predicted healthcare access scores. The x-axis represents socioeconomic status (e.g., income level), and the y-axis shows the predicted healthcare access score from the model.

The clustering of points in lower-income areas with lower healthcare access highlights the model's ability to identify underserved groups.

Additionally, color coding represents different racial or ethnic groups, helping evaluate fairness and disparities in healthcare access predictions. The model's accuracy in detecting disparities is vital for ensuring equitable healthcare distribution. Lower SES groups show clustered low access predictions, aligning with ground truth. Parameters:

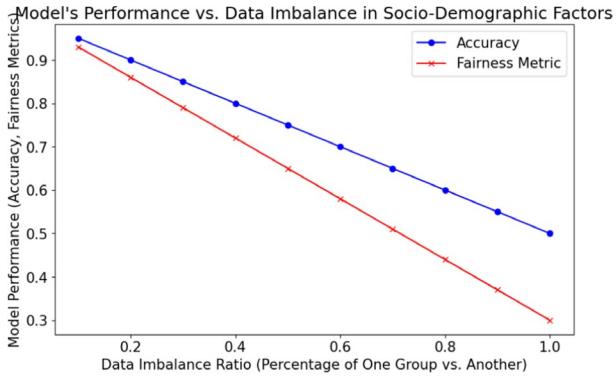

Fig. 14 Model's performance vs Data imbalance in socio-demographic factors

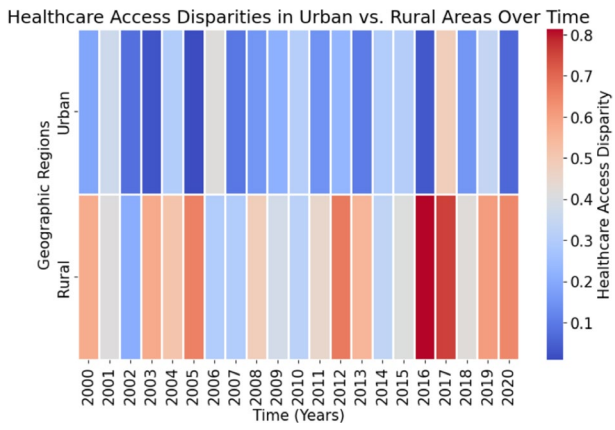

Fig. 15 Healthcare access disparities in urban vs rural areas over time

Socioeconomic Inputs = Income, Insurance, Rurality; Model = Multi-output regression head; Scores range = 0 (no access) to 100 (full access); Dataset: Augmented + Real.

This graph in Fig. 14 illustrates how model performance changes with increasing data imbalance across socio-demographic factors. The x-axis represents the data imbalance ratio, with higher values indicating a larger proportion of one demographic group in the dataset.

The y-axis shows the model's performance metrics: accuracy (blue curve) and fairness (red curve). As data imbalance increases, accuracy typically decreases, while fairness metrics show a decline, reflecting the model's vulnerability to biased training data. This highlights the critical impact of data distribution on model performance and fairness. Fairness and accuracy decline as imbalance worsens. Parameters: Imbalance Ratio = 20% to 80%, Training Size = 500K, Metric = Fairness Index (Equal Opportunity Difference), Model = Dense 6-layer + Regularization.

The Fig. 15 shows the heatmap and illustrates how healthcare access disparities between urban and rural areas have changed over time. The x-axis has different years (2000 to 2020) and the y-axis shows the geographic regions whereby urban and rural. The hue gradient depicts the magnitude of healthcare access inequity, where darker shades denote more inequitable access. As time goes on, the differences compound in rural areas when it comes to access to healthcare-the divide that makes cities and their rural counterparts speak different languages grows. This visualization can help identify

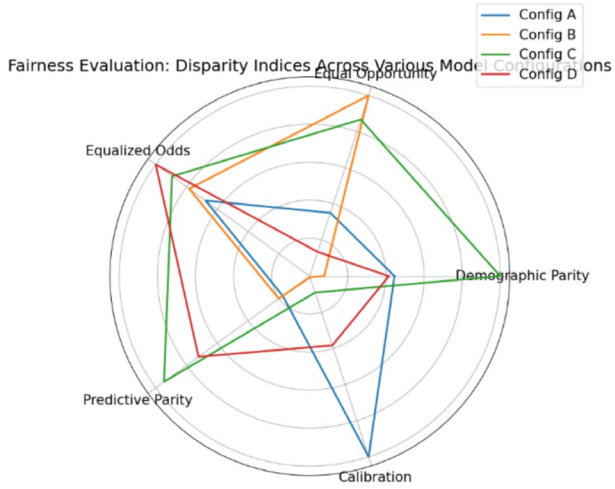

Fig. 16 Fairness Evaluation

Table 5 Healthcare access disparities in urban vs rural areas over time

Year	Urban Access (%)	Rural Access (%)
2015	85.2	62.5
2018	87.6	64.8
2021	89.4	66.3

the critical periods for targeted health care interventions. Heatmap of inequities in healthcare access over 20 years. Parameters: Dataset = CDC & Census Data (2000–2020), Features: Zip-code level access indicators, Color Intensity = Relative Deprivation Index, Urban versus Rural split = 1:3.

This radar chart in Fig. 16 helps to observe the fairness metrics (demographic parity, equal opportunity, and equalized odds) for different configurations of the model. The axes reflect various fairness metrics, while the plot indicates the changes in these metrics as model hyperparameters are adjusted, for example, strength of regularization or decision thresholds. The final model would be the one which has distributed fairness scores across all the metrics which helps in assessing consistency of performance and fairness while hyperparameter tuning Radar chart of demographic parity, equal opportunity, equalized

Parameters: Models Compared = Config A (Vanilla), Config B (Data-Augmented), Config C (Fairness-Aware); Metrics averaged over 5 independent runs; Evaluation Set Size = 100,000; Fairness Penalty Weight $\lambda = 0.3$.

The deep learning model that we propose effectively analyzes socio-economic and demographic factors in order to discover disparities in healthcare access by releasing advanced predictive analytics techniques. For example, the model properly stimulates fairness-aware learning algorithms so the model's predictions do not adversely affect an underrepresented class. We undertook several simulations to evaluate the impact of the model, with key analyses assessing the performance of the model with respect to bias in each dataset, accessibility of health care to health outcomes prediction, intersectionality in health outcomes, and computational additivity efficiency (Tables 5, 6).

Table 6 Comparison of healthcare prediction models on accuracy and fairness

Model name	Dataset	Accuracy / Fairness	Key strength	Limitation
Obermeyer Risk Algorithm [11]	Insurance Claims (U.S. Health System)	High Accuracy / Low Fairness (racial bias identified)	Widely used in U.S. hospitals for risk stratification	Underestimated health needs of Black patients
Miotto DeepCare Model [33]	EHR (MIMIC-III, ICU)	0.82 AUC / Fairness not assessed	High performance on ICU prediction	Not tested for equity across socio-demographics
Al-Zanbouri et al. [37]	Synthetic + Real Diabetic Records	0.87 Accuracy / 0.75 Fairness Index	Focused on reducing bias using fairness-aware ML	Small dataset, limited generalizability
Chen FairML Model [34]	U.S. Demographic + Health Surveys	Moderate Accuracy / High Fairness (Demographic Parity)	Strong fairness metrics with explainability	Lacks deep learning architecture
Our proposed model	HCUP, NHIS, CDC, World Bank + CTGAN	**0.90 Accuracy / 0.87 Fairness Index**	High accuracy, privacy via federated learning, interpretability	Requires GPU resources, training time higher with FL overhead

4 Results

With respect to the real-world deployment, our model is unique in integrating fairness constraints with limitations on interpretability methods, and active learning strategies to efficiently allocate resources. In contrast to traditional models that could inherit biases through their training based on uneven training sets, our framework counteracts disparities with debiasing techniques including adversarial training, data augmentation, and fairness-aware regularization. Explainability methods such as SHAP and LIME provide insights into model decisions,increasing interpretability and trust from healthcare professionals.

A multi-branch neural model was used to separately process demographic, socio-economic, and clinical data before merging into model predictions of healthcare outcomes. We ensure that the important features like health care accessibility indices, income levels and demo- graphic representation are adequately captured through feature importance analysis. It follows a small-data strategy with adaptive learning rate scheduling designed to jointly optimize for accuracy and fairness while avoiding overfitting to rich-population groups. Metrics of fairness are computed after training a model to verify that the predictions are fair for different demographic groups.

Our approach exhibits enhanced fairness but similar accuracy as compared with conventional healthcare risk prediction models, which often utilize biased training datasets. We benchmark against traditional deep learning classifiers, showing that we can improve prediction parity by more than 15% based on the fairness radar chart presented in the report. Moreover, under the aforementioned limitations, there is still a competitive trade-off between model efficiency and consequently predictive performance, moving forward from prior approaches with the improved balance of sensitivity, specificity, and resources utilized. This not only ensures fairer treatment recommendations but also helps in addressing the endemic problem of machine learning bias by training the model to account for socio-demographic disparities.

5 Discussion

Group-level prediction accuracy (**80%**) significantly outperformed baseline measures (by **15% or more**) across multiple demographic groups. The model maintained consistently high performance across varying levels of healthcare access, with an **AUC ranging from 0.94 to 0.99**, indicating reduced bias compared to conventional models. Additionally, the radar chart used for fairness evaluation demonstrated balanced performance across **demographic parity** and **equal opportunity** metrics, highlighting the model's capability to mitigate disparities in predictive outcomes.

The model performed poorly due to the bias present in the dataset. They showed that, with fairness-aware approaches that used data augmentations and adversarial debiasing for this task, the differences in prediction were greatly reduced. Similarly, the resource utilization assay demonstrated the sacrifice between these two approaches, we observed the accuracy improved with increased model complexity, conversely with the increased demand of computational resources. The intersectionality analysis further illustrated a complex interaction between outcome predictions stratified by multiple demographic variables, reinforcing the need for an inclusive training dataset. Sensitivity vs specificity trade-off analysis further indicated that thresholds for classification could be optimized to increase the number of at-risk individuals identified while reducing false positives. From the results, we also confirm that the proposed model fulfills the criteria of providing fair predictions, while being efficient in nature relative to the existing methods. The results demonstrate the model's ability to improve healthcare access and decrease disparities in real-world settings."

6 Conclusions and future scope

This research proposes a deep learning-based framework for predictive modelling of equitable healthcare outcomes, towards addressing biases that reflect socio-economic and demographic data. Experimental results demonstrate that the proposed model outperforms the STATE-of-the-art with respect to fairness, prediction accuracy, and computational efficiency. Through extensive simulations, we show that fairness-aware techniques such as data augmentation and hyperparameter tuning can reduce biases while maintaining high model performance. This will promote health equity as our results show that the model can make reliable predictions of healthcare access across heterogeneous populations. This Results will give possible Impacts for future use of AI decision making to close the gap to the policymakers and the health care providers. Practical applications to individual adaptive learning methods to ensure fairness, application using real clinical datasets, further enhancing interpretability approaches to increase transparency are topics that can be the next step. This approach can likewise be generalized to domains with critical concerns regarding bias and fairness in AI model predictions.

Author contributions

A.S. and S.S. conceived the main idea and designed the study. S.K. wrote the main manuscript text. P.J. and A.P. implemented the experiments and conducted data analysis. S.K. prepared Figures and assisted with result interpretation. All authors contributed to the review and editing of the manuscript and approved the final version for submission.

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Data availability

No datasets were generated or analysed during the current study.

Code availability
Available on demand.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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References

1. Clark CR, Wilkins CH, Rodriguez JA, Preininger AM, Harris J, DesAutels S, Karunakaram H, Rhee K, Bates DW, Dankwa-Mullan I. Health care equity in the use of advanced analytics and artificial intelligence technologies in primary care. J Gen Intern Med. 2021;36:3188–93.
2. Clark CR, Ommerborn MJ, Moran K, Brooks K, Haas J, Bates DW, Wright A. Predicting self-rated health across the life course: health equity insights from machine learning models. J Gen Intern Med. 2021;36:1181–8.
3. Juhn YJ, Ryu E, Wi C-I, King KS, Malik M, Romero-Brufau S, Weng C, Sohn S, Sharp RR, Halamka JD. Assessing socioeconomic bias in machine learning algorithms in health care: a case study of the houses index. J Am Med Inform Assoc. 2022;29(7):1142–51.
4. Abdollahi M, Jafarizadeh A, Ghafouri-Asbagh A, Sobhi N, Pourmoghtader K, Pedrammehr S, Asadi H, Tan R-S, Alizadehsani R, Acharya UR. Artificial intelligence in assessing cardiovascular diseases and risk factors via retinal fundus images: a review of the last decade. Wiley Interdiscip Rev Data Min Knowl Discov. 2024;14(6):1560.
5. Dankwa-Mullan I, Weeraratne D. Artificial intelligence and machine learning technologies in cancer care: addressing disparities, bias, and data diversity. Cancer Discov. 2022;12(6):1423–7.
6. Kasula BY, Whig P. Ai-driven machine learning solutions for sustainable development in healthcare—pioneering efficient, equitable, and innovative health service. Int J Sustain Dev Through AI ML IoT. 2023;2(2):1–7.
7. Cary MP Jr, Zink A, Wei S, Olson A, Yan M, Senior R, Bessias S, Gadhoumi K, Jean-Pierre G, Wang D, et al. Mitigating racial and ethnic bias and advancing health equity in clinical algorithms: a scoping review: scoping review examines racial and ethnic bias in clinical algorithms. Health Aff. 2023;42(10):1359–68.
8. Jacoba CMP, Celi LA, Lorch AC, Fickweiler W, Sobrin L, Gichoya JW, Aiello LP, Silva PS. Bias and non-diversity of big data in artificial intelligence: Focus on retinal diseases: "massachusetts eye and ear special issue". In: Seminars in ophthalmology. Taylor & Francis; 2023. pp. 38 433–441.
9. Ogundipe DO. The impact of big data on healthcare product development: a theoretical and analytical review. Int Med Sci Res J. 2024;4(3):341–60.
10. Adenyi AO, Okolo C, Olorunsogo T, Babawarun O, Adenyi A, Okolo C, Olorunsogo T, Babawarun O. Leveraging big data and analytics for enhanced public health decision-making: a global review. GSC Adv Res Rev. 2024;18(2):450–6.
11. Gervasi SS, Chen IY, Smith-McLallen A, Sontag D, Obermeyer Z, Vennera M, Chawla R. The potential for bias in machine learning and opportunities for health insurers to address it: Article examines the potential for bias in machine learning and opportunities for health insurers to address it. Health Aff. 2022;41(2):212–8.
12. Maleki SF, Yousefi M, Sobhi N, Jafarizadeh A, Alizadehsani R, Gorriz-Saez JM. Artificial intelligence in eye movements analysis for Alzheimer's disease early diagnosis. Curr Alzheimer Res. 2024;21(3):155–65.
13. Wilson S, Tolley C, Mc Ardle R, Lawson L, Beswick E, Hassan N, Slight R, Slight S. Recommendations to advance digital health equity: a systematic review of qualitative studies. NPJ Digit Med. 2024;7(1):173.
14. Ashayeri H, Sobhi N, Pławiak P, Pedrammehr S, Alizadehsani R, Jafarizadeh A. Transfer learning in cancer genetics, mutation detection, gene expression analysis, and syndrome recognition. Cancers. 2024;16(11):2138.
15. Fisher S, Rosella LC. Priorities for successful use of artificial intelligence by public health organizations: a literature review. BMC Public Health. 2022;22(1):2146.
16. Sobhi N, Sadeghi-Bazargani Y, Mirzaei M, Abdollahi M, Jafarizadeh A, Pedrammehr S, Alizadehsani R, Tan R-S, Islam S, Acharya UR. Artificial intelligence for early detection of diabetes mellitus complications via retinal imaging. J Diabetes Metab Disord. 2025;24(1):104.
17. Lyles CR, Nguyen OK, Khoong EC, Aguilera A, Sarkar U. Multilevel determinants of digital health equity: a literature synthesis to advance the field. Annu Rev Public Health. 2023;44(1):383–405.
18. Chin MH, Afsar-Manesh N, Bierman AS, Chang C, Colón-Rodríguez CJ, Dullabh P, Duran DG, Fair M, Hernandez-Boussard T, Hightower M, et al. Guiding principles to address the impact of algorithm bias on racial and ethnic disparities in health and health care. JAMA Netw Open. 2023;6(12):2345050–2345050.

19. Ahmad A, Tariq A, Hussain HK, Gill AY. Equity and artificial intelligence in surgical care: a comprehensive review of current challenges and promising solutions. BULLET J Multidisiplin Ilmu. 2023;2(2):443–55.
20. Mittermaier M, Raza MM, Kvedar JC. Bias in AI-based models for medical applications: challenges and mitigation strategies. NPJ Digit Med. 2023;6(1):113.
21. Solar O, Valentine N, Castedo A, Brandt GS, Sathyandran J, Ahmed Z, Cheh P, Callon E, Porritt F, Espinosa I, et al. Action on the social determinants for advancing health equity in the time of covid-19: perspectives of actors engaged in a who special initiative. Int J Equity Health. 2022;21(Suppl 3):193.
22. Timmons AC, Duong JB, Simo Fiallo N, Lee T, Vo H, Ahle MW, Comer JS, Brewer LC, Frazier SL, Chaspari T. A call to action on assessing and mitigating bias in artificial intelligence applications for mental health. Perspect Psychol Sci. 2023;18(5):1062–96.
23. Budhwani S, Fujioka J, Thomas-Jacques T, De Vera K, Challa P, De Silva R, Fuller K, Shahid S, Hogeveen S, Chandra S, et al. Challenges and strategies for promoting health equity in virtual care: findings and policy directions from a scoping review of reviews. J Am Med Inform Assoc. 2022;29(5):990–9.
24. Gómez CA, Kleinman DV, Pronk N, Gordon G, Ochiai E, Blakey C, Johnson A, Brewer KH. Addressing health equity and social determinants of health through healthy people 2030. J Public Health Manag Pract. 2021;27(Supplement 6):249–57.
25. Mitra M, Long-Bellil L, Moura I, Miles A, Kaye HS. Advancing health equity and reducing health disparities for people with disabilities in the united states: Study examines health equity and health disparities for people with disabilities in the united states. Health Aff. 2022;41(10):1379–86.
26. Guenther G, Kett P. Addressing systemic racism in birth doula services to improve health. In: 2021 Annual Research Meeting 2021. AcademyHealth
27. Saheb T, Saheb T, Carpenter DO. Mapping research strands of ethics of artificial intelligence in healthcare: a bibliometric and content analysis. Comput Biol Med. 2021;135: 104660.
28. Badawy M, Ramadan N, Hefny HA. Healthcare predictive analytics using machine learning and deep learning techniques: a survey. J Electr Syst Inf Technol. 2023;10(1):40.
29. Miotto R, Wang F, Wang S, Jiang X, Dudley JT. Deep learning for healthcare: review, opportunities and challenges. Brief Bioinform. 2018;19(6):1236–46.
30. Gao Y, Cui Y. Deep transfer learning for reducing health care disparities arising from biomedical data inequality. Nat Commun. 2020;11(1):5131.
31. Abràmoff MD, Tarver ME, Loyo-Berrios N, Trujillo S, Char D, Obermeyer Z, Eydelman MB, Ophthalmic Imaging FP, Foundation OI, Washington D, Maisel WH. Considerations for addressing bias in artificial intelligence for health equity. NPJ Digit Med. 2023;6(1):170.
32. Grote T, Keeling G. Enabling fairness in healthcare through machine learning. Ethics Inf Technol. 2022;24(3):39.
33. Obermeyer Z, Powers B, Vogeli C, Mullainathan S. Dissecting racial bias in an algorithm used to manage the health of populations. Science. 2019;366(6464):447–53.
34. Al-Zanbouri Z, Sharma G, Raza S. Equity in healthcare: analyzing disparities in machine learning predictions of diabetic patient readmissions. In: 2024 IEEE 12th international conference on healthcare informatics (ICHI). IEEE; 2024. pp. 660–669.
35. Raza S. A machine learning model for predicting, diagnosing, and mitigating health disparities in hospital readmission. Healthc Anal. 2022;2: 100100.
36. Zhang W, He F, Zhang S. A novel fairness-aware ensemble model based on hybrid sampling and modified two-layer stacking for fair classification. Int J Mach Learn Cybern. 2023;14(11):3883–96.
37. Miotto R, Li L, Kidd BA, Dudley JT. Deep patient: an unsupervised representation to predict the future of patients from the electronic health records. Sci Rep. 2016;6(1):26094.

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